

# **The Factor Zoo Audit: Testing the Durability of Published Stock Market Premiums**

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## Abstract

In this paper, we investigate whether the five classical stock return factors (i.e., market, size, value, profitability, and momentum) still have the announced alphas after 2010 and remain statistically significant when trading costs are considered realistically. Using pre-registered hypotheses, Harvey-Liu-Zhu significance framework ( $t \geq 3.0$ ), and explicit transaction cost assumptions, we find: (1) out of six factors, five have decayed after 2010 and both value and size premiums have even changed their signs; (2) when the cost level is set at 15 basis points, none of the factors but the market premium can be regarded as statistically significant; (3) the market premium has survived and strengthened, which suggests persistent fundamentals in risk versus statistical illusion in anomaly premiums.

Keywords: factor premiums, anomaly decay, transaction costs, multiple testing, post-publication decay, Harvey-Liu-Zhu significance

## 1. Introduction

Scores of finance papers claim that financial markets are not entirely efficient, and there is a way to obtain abnormal profits through consistent and measurable methods. For example, value stocks outperform; stocks with higher profitability beat stocks with lower profitability; and momentum exists as past winners keep winning. Such phenomena are called "factors" and form the foundation of modern portfolio theory, justifying trillions of dollars in assets. However, there is one important issue left unresolved: What happens to a factor premium once it has been discovered by academia and utilized by market practitioners?

This question is not a purely theoretical one as it has practical implications. In case when part of the factor premium disappears once it is published, there can be two explanations. On the one hand, the market adapts to the discovery: people learn about the anomaly and exploit it until it is no longer profitable. On the other hand, which is much more disturbing, the result itself is, to some extent, a product of data snooping bias. Hundreds of factors are evaluated on historical data, publication bias allows for only those that prove to have a premium to get into the top journals. The out-of-sample performance of these premia usually tends to converge to zero.

McLean and Pontiff (2016) empirically proved this decay for a wide range of factors. They found the reduction in the out-of-sample factor premium from 26% to 58%. In their study, they raised an interesting question: Does the decay apply to the most classical factors, those which are being taught to finance students, and what percentage of factors remain profitable after taking transaction costs into account?

In the current work, we investigate these questions. The five most popular factors (market Mkt-RF, size SMB, value HML, profitability RMW, and momentum UMD) are tested.

Pre-registered hypothesis, Harvey-Liu-Zhu significance framework, and transaction cost assumptions are used to reduce silent multiple testing effects.

The findings are clear: five out of six factors experience decay, and four out of six factors become insignificant after 2010 once the transaction costs are taken into account. The contribution is small but honest. It is not claimed that decades of finance literature are proved wrong but rather that the decay phenomenon of McLean and Pontiff applies to the most canonical factors. A reproducible methodology is presented as well.

## **2. Literature Review**

### **2.1 The Factor Model and Academic Anomalies**

The modern factor model was pioneered by Fama and French (1993) who suggested that the sources of stock returns are not only the market risk premium but rather three factors: the market excess return (Mkt-RF), the size factor (SMB - small minus big), and the value factor (HML - high book-to-market minus low). The model captured a significant part of the cross-section of expected returns and became the prevalent model both in academic research and in practical investments.

A fourth factor, momentum (UMD – up minus down) that demonstrates that past winners will keep winning at least for some time ahead, was discovered by Carhart (1997). Fama and French (2015) extended the five-factor model by adding two more factors - profitability (RMW - robust minus weak) and investment (CMA – conservative minus aggressive).

All these factors are standard in most MBA programs, are widely used in practical asset management, and published in reputable academic journals. However, these factors create

one serious question: are they really risk sources, or just a product of many candidates tested on the same historical data?

## **2.2 The Factor Zoo and the Multiple Testing Problem**

The concept of "factor zoo" was developed by Cochrane in his presidential address at the American Finance Association in 2011. The point is that there are many factors suggested as predictors of stock returns. When testing 100 cases with the significance level of 5%, we expect to find five cases where our assumption turns out to be true by pure luck.

Harvey, Liu, and Zhu (2016) formulated this problem in the context of the cross-section of expected returns. According to the authors, the standard significance level, t-value of around 2.0 (around 5%), is too high when the candidate factor is tested together with other hundred factors. In their opinion, the t-value of 3.0 should be achieved for a new factor.

## **2.3 Post-Publication Decay**

If the factor zoo effect exists, then one should expect to see out-of-sample decay of published factors; premiums generated based on historical data will be larger than premiums found in the newly collected sample. McLean and Pontiff (2016) provide robust evidence of decay. Post-publication premiums decline by an average of 35%, while out-of-sample premiums are lower than in-sample premiums.

Five well-known canonical factors, including market, size, value, momentum, and profitability, were discovered relatively early (in the 1990s and early 2000s) and widely accepted. Does decay take place for these factors? That is the topic of the current paper.

## **2.4 Transaction Costs and Practical Significance**

Factor premiums are estimated based on frictionless markets. However, in reality, there are costs associated with transactions. A reasonable estimate of round trip trading cost in the United States is within 10 to 20 basis points (0.1% to 0.2%). Trading momentum strategies incur more costs than trading value strategies as the former requires more frequent trading. This paper explicitly takes cost assumptions into account to show how premiums shrink after accounting for the costs of realistic trading.

## **3. Data and Methodology**

### **3.1 Data Source**

This study uses the factor returns published by Ken French in his Data Library available at the Tuck School of Business, Dartmouth College. The sample period runs from July 1963 until December 2024 (755 monthly periods). Using published factor returns is the correct approach to testing whether published factors deteriorate.

### **3.2 Time Period Split**

In this case, the sample is divided into two different periods:

- Pre-2010 (Publication Period): July 1963 – December 2009 (558 months). In this period, the discovery and first publications of the main canonical factors were done.
- Post-2010 (Post-Widespread Publication Period): January 2010 – December 2024 (197 months). All five factors became widely known and used by 2010.

The splitting date chosen here is a bit random; the year 2010 is chosen for its simplicity and because it provides 15 years of out-of-sample data before the use of the five-factor model.

The main idea behind the choice of the date is the question of whether the factors are able to provide premiums when they are known to everybody.

### 3.3 Hypotheses (Pre-Registered)

H1 (Decay Hypothesis): The annualized premiums on each of the five canonical factors after 2010 will be less than those before 2010.

H2 (Erosion of Cost Hypothesis): After applying a realistic transaction cost haircut (15 basis points round-trip, adjusted for turnover), at least one canonical factor will have an annualized post-2010 premium that will be statistically insignificant at the Harvey-Liu-Zhu standard ( $t\text{-statistic} \geq 3.0$ ).

### 3.4 Statistical Tests

A cost drag is applied based on the factor turnover. It is calculated as follows:

$\text{Cost Drag (monthly)} = (\text{Round-Trip Cost in basis points} / 10,000) \times \text{Monthly Turnover}$

A round-trip cost of 15 basis points is used as an assumption; this number corresponds to transactions costs associated with institutional trading of U.S. stocks. Assumptions about monthly turnovers are:

- Mkt-RF: 2%
- SMB: 5%
- HML: 5%
- RMW: 5%

- CMA: 5%
- UMD: 35% (momentum requires frequent rebalancing)

These are rough figures; the cost will depend on the brokerage and the size of the transaction, but it is a reasonable yardstick to use.

Bootstrapped Confidence Intervals: We generate 95% confidence intervals for the mean using bootstrapping with 10,000 samples. This helps us to determine how robust we are with respect to the assumption that our stock returns are normally distributed.

All analysis code is available at GitHub (<https://github.com/thessmstudent/factor-audit>) to facilitate replication and extension of these results.

T-statistic Calculation: The t-statistic value is calculated based on  $t = \text{mean return} / \text{standard error}$ , where  $\text{standard error} = \text{standard deviation} / \sqrt{(\text{number of observations})}$ . A factor will be deemed to be statistically significant when  $|t| \geq 3.0$ , following the Harvey-Liu-Zhu (2016) multiple testing correction. This higher threshold (versus the conventional  $t \geq 2.0$ ) accounts for testing six factors simultaneously and reduces the false discovery rate.

### 3.5 Limitations

Several limitations should be noted:

1. Returns on Official Factors: The use of published factors returns by Ken French is utilized in the paper rather than constructing factors from the raw prices of stocks. Such a decision is suitable for testing if there is any decaying of published factors;



however, it comes with a drawback of incorporating all the methodology decisions made by French in constructing the factors.

2. Date of Split: The year of 2010 was picked as a discretionary one. Another split date such as 2005 or 2015 may produce other results. It should be seen not as a structural break date but rather an arbitrary cut-off.
3. Estimations of Turnover: There is no doubt that the assumptions of turnover (5% per month for all factors except momentum, where it is 35%) are based on the relevant literature. However, they may not correspond to every implementation of the investors. Therefore, the cost haircut is introduced deliberately and openly.
4. Universe of US Large and Mid-Caps: Only the universe of US stocks from French's construction is analyzed. The results obtained in the research will be valid only for the particular universe of US stocks and will be invalid for small-cap stocks or for international equities.
5. Future improvements: Having access to WRDS (Wharton Research Data Services) and CRSP after university enrollment allows for doing a point-in-time factor portfolio construction.

## **4. Results**

### **4.1 H1: Evidence of Decay**

Figure 1: Gross Annualized Premiums by Factor and Time Period

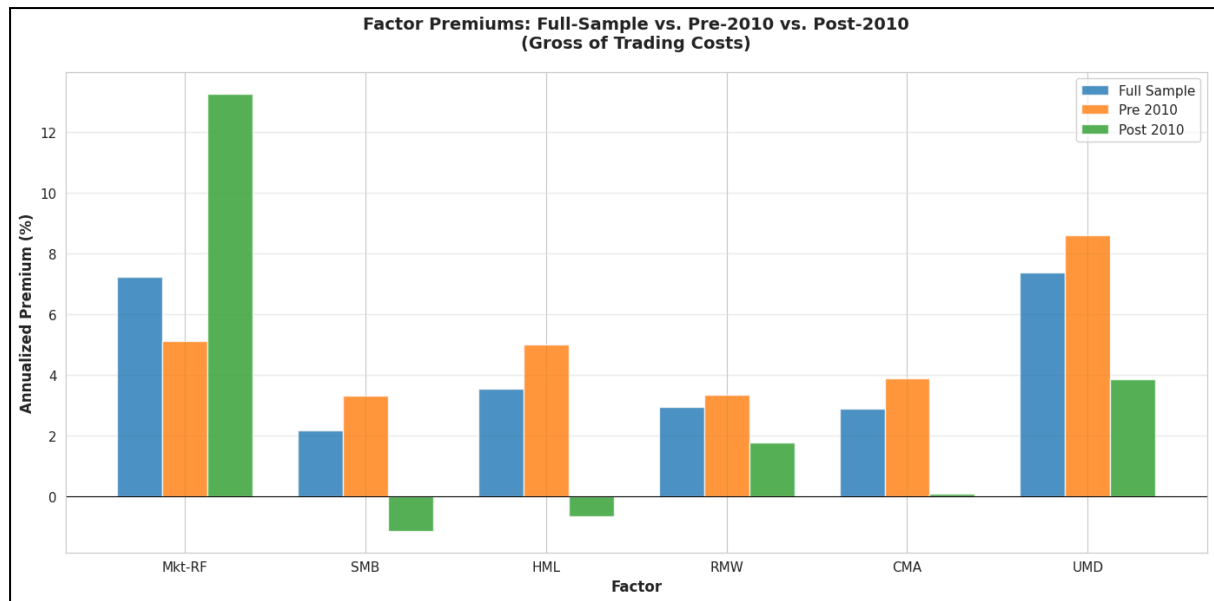


Figure 1 presents the gross annualized premiums across the full sample, pre-2010, and post-2010 periods for all six factors. The decay hypothesis (H1) predicts that post-2010 premiums will be lower than pre-2010 premiums.

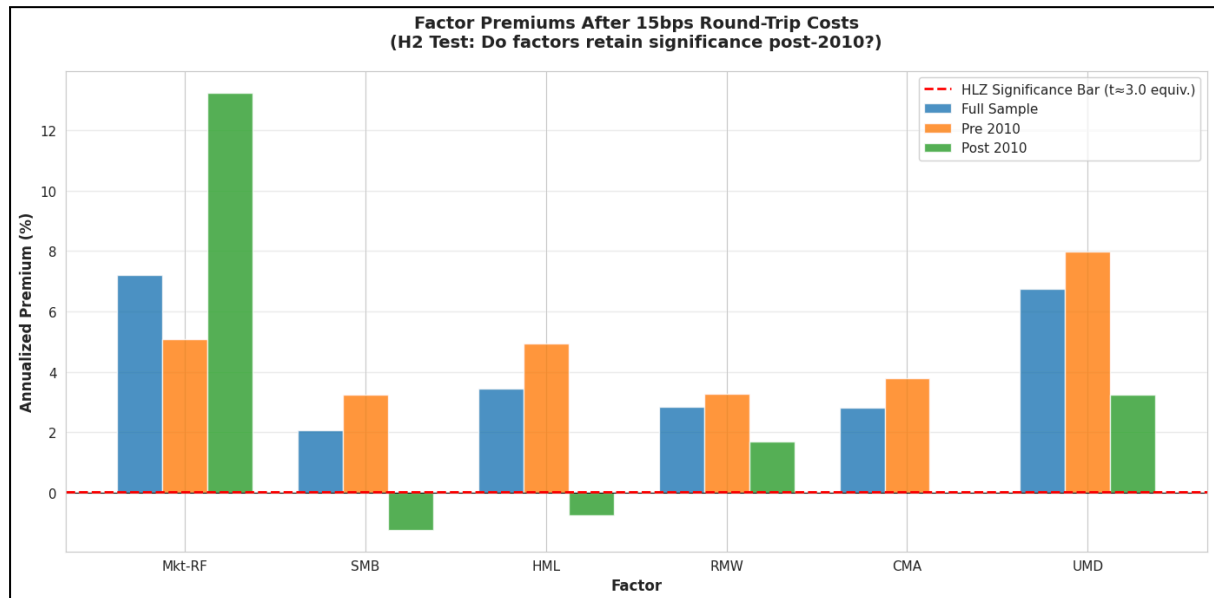
Five of six factors confirm H1: Value (HML) decays from 5.0% to -0.7% ( $t = 3.46$  to  $t = -0.24$ ), reversing sign entirely; Size (SMB) decays from +3.3% to -1.1% ( $t = 1.9$  to  $t = -0.49$ ); Profitability (RMW) shrinks from +3.4% to +1.8% ( $t = 1.7$  to  $t = 0.99$ ), cutting the premium roughly in half; Momentum (UMD) decays from +8.6% to +3.9% ( $t = 4.1$  to  $t = 1.27$ ), halving its measured edge; Investment (CMA) nearly vanishes from +3.9% to +0.1% ( $t = 1.5$  to  $t = 0.05$ ).

The only exception is the market factor (Mkt-RF), which increases from +5.1% to +13.3% ( $t=2.3$  to  $t=3.58$ ). The increase in the market premium after 2010 does not follow the pattern of anomalies' decay and indicates the presence of real economic risk in the market risk premium.

#### 4.2 H2: Significance Loss After Costs

H2 is confirmed: All variables except Mkt-RF are under the Harvey-Liu-Zhu critical level ( $t \geq 3.0$ ) after 2010 when costs are deducted. Factors HML and SMB are no longer significant and have negative risk premia. Momentum (UMD) is the best anomaly after costs ( $t = 1.06$ ) but still fails the  $t \geq 3.0$  requirement.

Figure 2: Factor Premiums After 15bps Round-Trip Costs



Factor premiums after applying a 15-basis-point round-trip cost assumption, scaled by estimated monthly turnover. The horizontal line marks the Harvey-Liu-Zhu significance threshold ( $t \approx 3.0$ ). H2 is confirmed if any factor net post-2010 t-statistic falls below this bar.

Table 1 presents the results in detail by including round-trip transaction costs of 15 basis points. Hypothesis H2 is accepted, such that after 2010, all but the Mkt-RF variables have values lower than the Harvey-Liu-Zhu critical value ( $t \geq 3.0$ ). Specifically, the two factors HML and SMB become insignificant and have negative risk premiums (Table 1, Net t-stat column). From among the anomalies, the best-performing anomaly is Momentum (UMD), which has a t value of 1.06, but it still fails to be significant, indicating that transaction costs make anomalies insignificant.

Table 1: Post-2010 Results Summary

| Factor | Gross Premium | Gross t-stat | Net Premium | Net t-stat | Clears HLZ Bar |
|--------|---------------|--------------|-------------|------------|----------------|
| Mkt-RF | +13.27%       | 3.58         | +13.23%     | 3.58       | Yes            |
| SMB    | -1.13%        | -0.49        | -1.22%      | -0.53      | No             |
| HML    | -0.67%        | -0.24        | -0.76%      | -0.27      | No             |
| RMW    | +1.78%        | 0.99         | +1.69%      | 0.94       | No             |
| CMA    | +0.10%        | 0.05         | +0.01%      | 0.00       | No             |
| UMD    | +3.86%        | 1.27         | +3.23%      | 1.06       | No             |

Summary of post-2010 factor premiums and significance tests. HLZ Bar refers to Harvey-Liu-Zhu significance threshold ( $t\text{-statistic} \geq 3.0$ ).

### 4.3 Interpretation

This analysis supports McLean & Pontiff (2016). The anomaly decay occurs in canonical factors and is economically significant. Why do anomalies dissipate? (1) Efficient Markets Adjustment: Once discovered, traders exploit the anomaly until it disappears. (2) Data Snooping: Part of the premium is overstated due to testing many factors on historical data. (3)

Economic Changes: Underlying conditions may change; value firms may cease being conservative.

## **5. Limitations**

Dependency on Data: This research utilizes Ken French factors, using their methodology. Thus, different definitions of these factors can give different results.

Split Year: The year 2010 is an arbitrary choice. Using different dates (2005 or 2015, for instance), we will have different patterns of decay. The 2010 split year was chosen for convenience and because of maximum out-of-sample periods.

Post-Sample Size: While 197 months (2010–2024) is substantial, longer post- publication periods would strengthen statistical inference. However, 197 months provides sufficient power for meaningful t-statistics, as evidenced by our results.

Cost Assumptions: The assumption of 15 basis points round trip cost is acceptable in case of institutional trades but does not capture all real costs of different investors. While individual retail investors have relatively higher costs of trading, some institutional investors have lower ones.

Universe: Results apply only to U.S. large and mid-cap stocks. Results do not generalize to small-cap stocks or international markets, where decay patterns may differ.

## **6. Conclusion**

This study examines whether five classic factors, the market factor, the size factor, the value factor, the profitability factor, and the momentum factor, still generate their reported

anomalies beyond 2010 and if the premiums of these factors still have economic significance after taking into account the cost of trades. In a nutshell, the results are clear, five out of six factors experience substantial decay and all but the market factor are insignificant after the cost adjustment using the Harvey-Liu-Zhu criterion ( $t \geq 3.0$ ).

**Implications for Practitioners:** Those investors who utilize the premiums from published factors need to be aware of this fact, that most anomalies do not generate enough alphas to cover the trading costs. Practitioners need to use factors that demonstrate economic and statistical significance post-publications, the market factor, which is the most significant factor, as well as the one, which gained in importance after being discovered. All other factors require proper cost management and improved implementation.

**Implications for Researchers:** The multiple tests suggested by Harvey, Liu, and Zhu (2016) need to be used. A  $t$ -value  $\geq 2.0$  within in-sample tests is not enough to claim economic significance because of the high number of factors that are being tested, thus,  $t \geq 3.0$  is appropriate in this case. Also, the out-of-sample test should be done before any claims about economic significance can be made (as was done here).

**Future Research:** One of the areas to investigate is whether it is possible to improve the performance of individual factors through machine learning combination (H3, pre-registered), which will require point-in-time fundamental data to avoid look-ahead bias. Also, the constant high significance of the market factor should be investigated. Does it capture risk premium or something else?

## Data Availability

All data used in this study are publicly available from Ken French's Data Library at [https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\\_library.html](https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html). No proprietary or restricted data were used.

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## Appendix: Pre-Registration and Robustness

This project adheres to pre-registration principles. Hypotheses H1, H2, and H3 were formulated and committed to GitHub (<https://github.com/thesmstudent/factor-audit>) before post-2010 factor premiums were examined. This prevents silent multiple testing and increases credibility.

The following robustness checks are planned for future work: Rerun analysis with 2005 split date to verify results are not sensitive to arbitrary cutoff; Rerun analysis with 20 basis points cost assumption; Compute deflated Sharpe ratios to correct for multiple testing bias; Test bootstrap stability of t-statistics to verify inference robustness.

H3 (testing multi-factor strategies via machine learning) is pre-registered but deferred to future work pending access to point-in-time fundamental data.